

TOUL ROSIERES, France

WMO: 071790

Lat: 48.783N

Lon: 5.983E

Elev: 938

StdP: 14.20

Time Zone: 1.00 (EUC)

Period: 10-19

WBAN: 34051

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	20.9	24.7	12.2	10.6	30.8	17.7	13.9	29.6	21.1	49.3	18.8	43.1	3.7	340	0.479

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.2	92.8	68.7	88.1	67.8	84.6	66.5	71.1	86.2	69.7	84.1	68.0	82.0	7.1	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.3	100.5	77.0	64.6	94.2	75.1	62.9	88.8	72.9	35.6	86.6	34.1	84.2	32.7	82.8	82.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.5	14.3	12.5	DB	16.2	98.4	6.0	3.6	11.9	101.0	8.3	103.2	5.0	105.2	0.6	107.9	(4)
(5)			WB	15.5	74.8	6.0	2.8	11.2	76.9	7.7	78.6	4.3	80.2	0.0	82.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	53.4	39.9	39.9	47.1	51.6	56.2	63.7	67.9	67.1	62.0	55.4	46.6	41.9		
	DBStd	11.92	6.15	7.36	5.32	6.30	6.61	7.77	6.74	7.04	6.15	6.93	6.97	6.28		
	HDD50	1233	315	286	120	56	16	3	0	0	2	31	146	258		
	HDD65	4644	778	703	554	404	285	114	48	57	129	304	551	717		
	CDD50	2462	3	3	31	103	208	416	555	531	361	198	46	7		
	CDD65	393	0	0	0	1	11	76	138	123	38	6	0	0		
	CDH74	5550	0	0	2	66	216	1071	1840	1719	579	56	1	0		
	CDH80	2137	0	0	0	6	43	417	743	759	164	5	0	0		
(14) Wind	WSAvg	5.2	5.0	5.7	5.9	6.0	5.8	5.2	5.1	4.7	4.4	4.8	5.2	4.8		
(15) Precipitation	PrecAvg	30.3	2.8	2.4	1.8	2.0	2.6	2.7	2.4	2.3	2.4	2.7	3.1	3.1		
	PrecMax	38.0	5.6	5.5	4.4	4.8	5.0	5.6	4.2	4.4	5.3	5.9	7.5	8.3		
	PrecMin	23.6	0.5	0.5	0.2	0.3	0.2	1.0	0.7	0.6	0.9	1.3	0.6	0.7		
	PrecStd	3.6	1.3	1.5	1.1	1.1	1.1	1.4	0.8	1.2	1.3	1.2	1.7	1.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.7	66.4	73.4	80.7	84.6	96.8	98.7	96.8	89.2	79.1	69.4	60.9		
		MCWB	51.4	50.1	53.1	58.7	65.5	69.4	68.7	68.8	66.1	63.2	58.3	52.9		
	2%	DB	55.0	59.3	68.4	75.2	80.4	89.2	91.5	92.9	84.5	75.1	64.5	57.5		
		MCWB	50.4	49.3	52.0	57.3	62.1	70.1	68.9	68.2	64.1	62.1	56.0	51.1		
	5%	DB	52.0	55.3	64.4	71.2	75.2	84.5	87.5	88.2	80.7	71.6	61.0	55.1		
		MCWB	48.9	47.7	50.8	54.3	60.8	67.2	68.3	67.5	63.2	60.6	54.7	50.4		
	10%	DB	49.8	51.7	59.3	66.2	71.4	80.5	84.1	83.9	76.9	67.9	57.6	52.1		
		MCWB	47.2	46.6	49.0	52.6	57.9	65.2	66.7	66.7	62.0	59.4	53.8	48.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.2	52.8	55.4	60.4	66.4	76.0	72.8	72.5	68.4	65.5	59.8	54.8		
		MCDB	55.7	60.6	67.3	75.2	81.6	88.7	88.5	88.1	82.2	73.7	66.4	58.1		
	2%	WB	51.4	50.7	53.3	58.0	64.2	71.0	71.0	70.5	66.7	63.6	57.6	53.0		
		MCDB	54.2	56.6	64.0	72.4	76.7	85.1	86.0	85.9	79.9	70.9	62.7	56.3		
	5%	WB	49.6	49.0	51.7	55.7	62.1	68.3	69.5	69.2	65.0	62.2	56.1	51.4		
		MCDB	51.9	53.7	61.2	67.7	72.0	82.5	83.9	84.0	76.3	68.9	60.2	54.6		
	10%	WB	47.4	47.0	50.2	53.9	59.8	66.2	67.7	67.5	63.2	60.4	54.2	49.5		
		MCDB	49.1	51.3	58.3	63.9	68.8	78.0	81.3	81.3	73.4	67.0	58.3	52.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	13.0	17.4	20.3	22.3	22.9	24.0	25.2	26.4	24.1	20.5	15.3	15.4		
		MCDBR	15.7	26.1	32.3	30.7	30.4	31.2	31.4	34.2	31.0	25.0	20.9	20.7		
		MCWBR	11.6	16.6	17.4	15.3	15.0	14.4	12.8	13.0	13.3	13.7	13.1	14.1		
	5% WB	MCDBR	13.6	20.1	25.0	27.6	26.6	28.7	28.2	29.2	26.0	20.5	17.4	15.1		
		MCWBR	11.0	14.2	14.6	14.6	14.7	14.7	13.4	13.3	13.0	12.1	11.8	11.3		
(40) Clear-Sky Solar Irradiance	taub		0.310	0.334	0.361	0.390	0.400	0.404	0.399	0.396	0.370	0.355	0.326	0.306		
	taud		2.439	2.386	2.323	2.258	2.267	2.286	2.292	2.328	2.376	2.408	2.437	2.421		
	Ebn at Noon		239	258	269	272	273	272	272	267	263	246	229	223		
	Edh at Noon		20	27	34	40	41	41	40	36	32	26	20	18		
(44) All-Sky Solar Radiation	RadAvg		281	521	907	1365	1614	1824	1721	1484	1137	671	326	225		
	RadStd		44	90	131	177	182	162	166	156	114	52	33	40		

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (46 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page